



TECHNICAL INFORMATION SHEET 31 - 2014

GAS CLASSIFICATION CHANGES UNDER ISO 10156

Background

When filling compressed gases, gas companies follow national and international standards to determine the correct product classification and identification. A gas classified as non-flammable will have a different transport label to a flammable gas. In some countries, regulations regarding the cylinder valve connection, as well as the cylinder shoulder colour are also used to differentiate a flammable gas from a non-flammable gas.

ISO 10156: 2010

ISO 10156, *Gases and gas mixtures - Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets*, defines a standard method for determining the flammable or oxidising potential of a gas or gas mixture. ISO 10156 was revised and a new version published in 2010. There were no changes to the classification of pure gases under this revision but there were changes that affect the classifications of various mixtures.

NOTE: ISO 10156 provides a calculation method for determining whether a gas or gas mixture is flammable in air.

New Tci Values

The Threshold combustion index (Tci) is defined as the maximum content of a flammable gas which, when mixed with nitrogen, is not flammable in air. Tci is also used as a reference parameter in any kind of mixture that contains a flammable component. The new standard includes changes to these Tci values for a number of gases. Within ISO 10156: 2010, new Tci values have been assigned to a variety of existing mixtures. Depending on the mixture the Tci value may be higher (less restrictive) or lower (more restrictive) than in the previous version of the standard.

Direct implications

Products that have been affected will see changes in the following areas:

- A new transportation identification label may be required.
- In some countries, a new cylinder shoulder colour may be required to indicate the change from either a non-flammable to a flammable mixture, or from a flammable to a non-flammable mixture.
- The Safety Data Sheet will need to be updated to include the changes for cylinder safety and transport.
- In some countries, a different cylinder valve outlet may be required.

Indirect implications for the user

- Storage conditions may need to be reviewed.
- Transport conditions may need to be reviewed.
- Existing risk assessments may need to be reviewed or new risk assessments conducted.

- Gas control equipment and supply system compatibility may need to be checked, as changes may be required for both cylinder connections and supply line labelling.
- A roll out programme may be needed to change over equipment and cylinders.

Some examples

Gas	Tci ISO 10156: 1996	Tci ISO 10156: 2010
Hydrogen (H ₂)	5.7	5.5
Methane (CH ₄)	14.3	8.7
Ethylene (C ₂ H ₄)	6.0	4.1
Ethylene Oxide (C ₂ H ₄ O)	3.1	4.8
R152a	4.6	8.7

A mixture with 7.5 % methane in argon was classified as non-flammable according to ISO 10156:1996, but became flammable according to ISO 10156: 2010. This will require a different UN number and proper shipping name, i.e. from UN1956 to UN1954, and where required, the shoulder colour will change from green to red. As noted, a new cylinder valve designation may also be required.

A mixture with 4 % ethylene oxide in nitrogen was classified as flammable according to ISO 10156: 1996, it became non-flammable according to ISO 10156: 2010. This will require a different UN number and proper shipping name, i.e. from UN1954 to UN1956, and where required, the shoulder colour will change from red to green. As noted, a new cylinder valve designation may also be required.

In both cases there will be changes required to the Safety Data Sheet.

Further information on the colour codes used to identify gas cylinders is available in BCGA TIS 6, *Cylinder identification. Colour coding and labelling requirements*.

References:

- 1) ISO 10156: 1996, Gases and gas mixtures. Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets.
- 2) BS EN ISO 10156-2: 2005. Gas cylinders. Gases and gas mixtures. Determination of oxidizing ability of toxic and corrosive gases and gas mixtures.
- 3) BS EN ISO 10156: 2010, Gases and gas mixtures. Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets.
- 4) BCGA TIS 6, Cylinder identification. Colour coding and labelling requirements.

For more information:

British Standards Institute (BSI)

www.bsigroup.co.uk

British Compressed Gases Association (BCGA)

www.bcgga.co.uk

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